

# CYBER SECURITY -TASK 3

## CYBER THREAT IDENTIFICATION

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### 1. INTRODUCTION

Cyber threats are malicious activities or attempts designed to gain unauthorized access to computer systems, networks, applications, or sensitive information. Cybercriminals use different techniques to steal data, damage systems, disrupt services, or manipulate users.

Identifying cyber threats is an important part of cybersecurity because recognizing a threat early can help prevent data loss, financial loss, privacy violations, and system damage.

Some of the most common cybersecurity threats include:

- Phishing
- Malware
- Ransomware
- Social Engineering
- Credential Theft
- Denial-of-Service (DoS/DDoS)
- Man-in-the-Middle (MITM) attacks
- SQL Injection
- Insider Threats
- Zero-Day Attacks

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### 2. OBJECTIVE

The main objective of this task is to recognize and understand common cybersecurity threats and identify how they affect users, systems, networks, and organizations.

Specific Objectives

1. Understand different types of cyber threats.
  2. Identify the targets of common cyber-attacks.
  3. Understand the possible impact of each threat.
  4. Learn basic methods of preventing cybersecurity attacks.
  5. Develop the ability to classify and identify cyber threats.
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### 3. COMMON CYBERSECURITY THREATS

#### 3.1 Phishing

Definition:

Phishing is a cyber-attack in which an attacker pretends to be a trusted person, company, or organization to trick users into providing sensitive information.

Attackers commonly use fake emails, messages, websites, or login pages.

Example:

A user receives an email that appears to come from a bank asking them to click a link and verify their account password. The link leads to a fake website controlled by the attacker.

Target:

- Users
- Email accounts
- Banking accounts
- Login credentials
- Organizations

Impact:

- Password theft
- Financial loss
- Identity theft
- Unauthorized account access
- Malware infection

Prevention:

- Do not click suspicious links.
  - Verify the sender before responding.
  - Check the website address carefully.
  - Use multi-factor authentication (MFA).
  - Avoid entering passwords on suspicious websites.
  - Report suspicious emails and messages.
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## 3.2 Malware

Definition:

Malware stands for Malicious Software. It refers to software intentionally created to damage systems, steal information, disrupt operations, or gain unauthorized access.

Common types of malwares include:

- Viruses
- Worms
- Trojans
- Spyware
- Adware
- Keyloggers

Example:

A user downloads an unknown application from an untrusted website. The application secretly installs malicious software on the computer.

Target:

- Computers
- Mobile devices
- Servers
- Networks

- User data

Impact:

- Data theft
- System damage
- Unauthorized access
- Loss of files
- Performance problems

Prevention:

- Install software only from trusted sources.
  - Keep operating systems updated.
  - Use reputable security software.
  - Avoid suspicious email attachments.
  - Scan external storage devices before opening files.
  - Maintain regular backups.
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### 3.3 Ransomware

Definition:

Ransomware is a type of malware that encrypts files or prevents access to systems and demands payment from the victim in exchange for restoring access.

Example:

An employee opens a malicious attachment, causing ransomware to encrypt important company files. A message then demands payment to restore access.

Target:

- Individuals
- Businesses
- Hospitals
- Government organizations

- Servers and databases

Impact:

- Files become inaccessible.
- Business operations may stop.
- Financial losses may occur.
- Sensitive information may be exposed.
- Recovery can require significant time and resources.

Prevention:

- Maintain regular offline or protected backups.
  - Keep systems and applications updated.
  - Avoid suspicious attachments and links.
  - Use endpoint security solutions.
  - Apply the principle of least privilege.
  - Train employees to recognize phishing attempts.
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### 3.4 Social Engineering

Definition:

Social engineering is a technique in which attackers manipulate people into revealing confidential information or performing actions that compromise security.

Instead of directly attacking a technical system, attackers exploit human trust and behavior.

Example:

An attacker calls an employee pretending to be an IT support person and asks for their login credentials.

Target:

- Employees
- Customers

- Administrators
- Organizations

Impact:

- Credential theft
- Unauthorized access
- Financial fraud
- Data leakage
- Compromise of company systems

Prevention:

- Never share passwords or OTPs.
- Verify unusual requests independently.
- Follow organizational security procedures.
- Provide cybersecurity awareness training.
- Be cautious with unexpected calls, emails, and messages.

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### 3.5 Credential Theft

Definition:

Credential theft is the unauthorized acquisition of usernames, passwords, authentication tokens, or other information used to access accounts.

Attackers may obtain credentials through phishing, malware, password attacks, or data breaches.

Example:

An attacker obtains a user's password through a fake login page and uses it to access the user's email account.

Target:

- Email accounts
- Banking accounts
- Social media accounts

- Corporate accounts
- Cloud services

Impact:

- Account takeover
- Data theft
- Identity theft
- Unauthorized transactions
- Further attacks on an organization

Prevention:

- Use strong and unique passwords.
- Enable MFA.
- Use a password manager.
- Never reuse passwords across important accounts.
- Be cautious of suspicious login pages.
- Change compromised passwords immediately.

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### 3.6 Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS)

Definition:

A Denial-of-Service (DoS) attack attempts to make a service or system unavailable by overwhelming it with excessive requests or traffic.

A Distributed Denial-of-Service (DDoS) attack uses multiple compromised devices or systems to generate traffic toward the target.

Target:

- Websites
- Web servers
- Online applications
- Network infrastructure

- Online services

Impact:

- Website becomes unavailable.
- Services become slow.
- Business operations may be interrupted.
- Customers may be unable to access services.
- Organizations may experience financial losses.

Prevention:

- Use firewalls and network security controls.
  - Use DDoS protection services.
  - Monitor network traffic.
  - Configure rate limiting.
  - Maintain redundant infrastructure.
  - Develop an incident response plan.
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### 3.7 Man-in-the-Middle (MITM) Attack

Definition:

A Man-in-the-Middle attack occurs when an attacker secretly intercepts or manipulates communication between two parties.

The attacker attempts to position themselves between the communicating parties without their knowledge.

Example:

An attacker on an unsecured network intercepts communication between a user and an online service.

Target:

- Network communication
- Login sessions
- Online transactions

- Users connected to insecure networks

Impact:

- Data interception
- Credential theft
- Session hijacking
- Modification of transmitted information
- Privacy loss

Prevention:

- Use HTTPS websites.
  - Avoid sensitive activities on unsecured public Wi-Fi.
  - Use secure and trusted networks.
  - Use VPNs where appropriate.
  - Keep devices and browsers updated.
  - Use MFA.
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### 3.8 SQL Injection

Definition:

SQL Injection is a web application attack in which malicious SQL input is inserted into application queries to manipulate the database.

It usually occurs when an application does not properly validate or parameterize user input.

Target:

- Web applications
- Databases
- Websites
- Database servers

Impact:

- Unauthorized access to database information
- Data modification
- Data deletion
- Exposure of sensitive information
- Potential compromise of application functionality

Prevention:

- Use parameterized queries or prepared statements.
  - Validate and sanitize input.
  - Apply least-privilege access to databases.
  - Use secure coding practices.
  - Conduct regular security testing.
  - Keep database software updated.
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### 3.9 Insider Threat

Definition:

An insider threat occurs when a person who has legitimate access to an organization's systems or information misuses that access, either intentionally or accidentally.

Insiders may include employees, contractors, or other authorized users.

Example:

An employee intentionally copies confidential company information to an unauthorized storage device.

Target:

- Company data
- Internal systems
- Databases
- Intellectual property
- Customer information

Impact:

- Data leakage
- Financial loss
- Privacy violations
- Intellectual property theft
- Damage to organizational operations

Prevention:

- Apply least-privilege access.
  - Monitor access to sensitive information.
  - Use strong authentication.
  - Maintain audit logs.
  - Conduct employee security awareness training.
  - Immediately revoke access when employees leave the organization.
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### 3.10 Zero-Day Attack

Definition:

A zero-day attack exploits a previously unknown or unpatched vulnerability in software or hardware.

The term "zero-day" refers to the fact that the vendor or security community may have had little or no time to develop and deploy a fix before exploitation occurs.

Target:

- Operating systems
- Applications
- Web browsers
- Network devices
- Enterprise software

Impact:

- Unauthorized access
- Data theft
- Malware installation
- System compromise
- Service disruption

Prevention:

- Keep software updated as soon as security patches become available.
- Use endpoint security solutions.
- Monitor security advisories.
- Use network segmentation.
- Apply least privilege.
- Maintain backups and incident response procedures.

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#### 4. THREAT CLASSIFICATION TABLE

| Threat     | Target                                   | Impact                                              | Prevention                                                |
|------------|------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------|
| Phishing   | Users, accounts, organizations           | Credential theft, fraud, malware infection          | Verify emails, avoid suspicious links, use MFA            |
| Malware    | Computers, mobiles, servers              | Data theft, system damage, unauthorized access      | Antivirus/security software, updates, trusted downloads   |
| Ransomware | Files, computers, servers, organizations | Data encryption, service disruption, financial loss | Backups, updates, security awareness, endpoint protection |

|                    |                                    |                                              |                                                                 |
|--------------------|------------------------------------|----------------------------------------------|-----------------------------------------------------------------|
| Social Engineering | Employees, customers, users        | Credential theft, fraud, unauthorized access | Awareness training, verify requests, never share credentials    |
| Credential Theft   | User and organizational accounts   | Account takeover, data theft                 | Strong passwords, MFA, password manager                         |
| DoS/DDoS           | Websites, servers, online services | Service unavailability and disruption        | DDoS protection, traffic monitoring, rate limiting              |
| MITM               | Network communication              | Data interception, credential theft          | HTTPS, secure networks, VPN where appropriate                   |
| SQL Injection      | Web applications and databases     | Data theft, modification or deletion         | Parameterized queries, input validation, secure coding          |
| Insider Threat     | Organizational data and systems    | Data leakage, fraud, information theft       | Least privilege, monitoring, access controls                    |
| Zero-Day Attack    | Vulnerable software and systems    | System compromise, data theft                | Security updates, monitoring, segmentation, endpoint protection |

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## 5. COMPARISON OF COMMON CYBER THREATS

| Threat   | Main Method Used                   | Human Interaction Required? |
|----------|------------------------------------|-----------------------------|
| Phishing | Fake emails, messages and websites | Usually yes                 |
| Malware  | Malicious software                 | Sometimes                   |

|                    |                                            |                                                  |
|--------------------|--------------------------------------------|--------------------------------------------------|
| Ransomware         | Malware that blocks/encrypts access        | Often                                            |
| Social Engineering | Psychological manipulation                 | Yes                                              |
| Credential Theft   | Stealing authentication information        | Often                                            |
| DoS/DDoS           | Flooding a service with traffic            | No direct interaction required                   |
| MITM               | Intercepting communication                 | Usually no direct interaction                    |
| SQL Injection      | Malicious database input                   | Usually no direct interaction after exploitation |
| Insider Threat     | Misuse of authorized access                | Yes, by an insider                               |
| Zero-Day           | Exploiting unknown/unpatched vulnerability | Not necessarily                                  |

## 6. BASIC CYBERSECURITY PREVENTION PRACTICES

Individuals and organizations can reduce cybersecurity risks by following basic security practices.

### 1. Use Strong Passwords

Passwords should be long, unique, and difficult to guess. Important accounts should not share the same password.

### 2. Enable Multi-Factor Authentication

MFA provides an additional layer of security by requiring another authentication factor besides the password.

### 3. Keep Software Updated

Operating systems, applications, browsers, and security software should be regularly updated to address known vulnerabilities.

#### 4. Avoid Suspicious Links and Attachments

Users should verify unexpected emails, messages, links, and attachments before opening them.

#### 5. Maintain Regular Backups

Important files should be backed up regularly. Protected or offline backups can help with recovery from ransomware and other incidents.

#### 6. Use Security Software

Antivirus and endpoint security solutions can help detect and prevent malicious activity.

#### 7. Follow Least Privilege

Users should receive only the permissions required to perform their work.

#### 8. Secure Network Connections

Sensitive activities should be performed using trusted and secure networks. HTTPS should be used for web communication.

#### 9. Conduct Security Awareness Training

Employees should be trained to recognize phishing, social engineering, suspicious links, and other common threats.

#### 10. Monitor Systems

Organizations should monitor network and system activity to identify unusual behavior and potential security incidents.